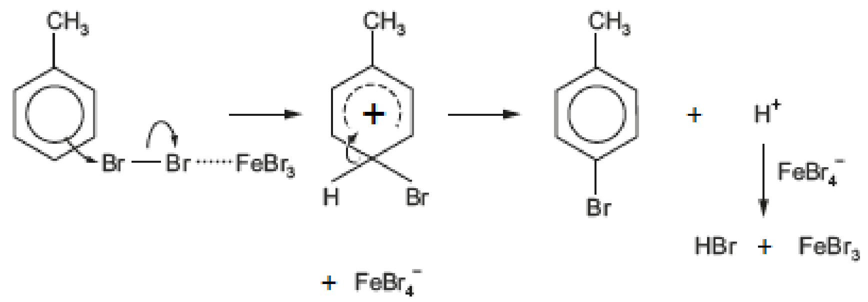
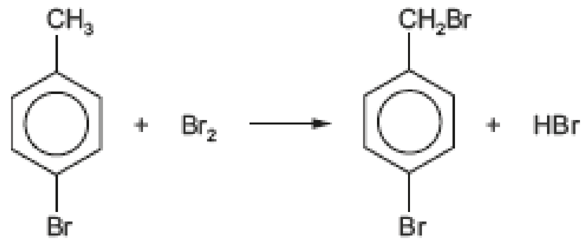
Question				Marking details	Marks available					
					AO1	**AO2**	**AO3**	**Total**	**Maths**	**Prac**
13.	(a)	(i)		it is compound **N** (1) singlet for the —CH_3 group bonded to the carbon atom of the double bond that is also bonded to the bromine atom (2.2δ) (1) the CH proton is as a quartet – split by the adjacent CH_3 protons (5.7δ) (1) the other —CH_3 group is a doublet, split by the adjacent CH proton (1.7δ) (1)			4	4		
		(ii)	I	add aqueous bromine / aqueous acidified manganate(VII) – this is decolourised by **L**, **M** and **N** but unaffected by bromocyclobutane		1		1		1
			II	bromocyclobutane would only give three ^{13}C signals (1) **L**, **M** and **N** would each give four signals as each carbon atom is in a different environment for these alkenes (1) accept answer based on C=C at δ 90 to 150 present in **L**, **M** and **N** but not in bromocyclobutane		2		2		
	(b)			 curly arrows (1) partial / full charges (1) regeneration of catalyst (1)	3			3		

Question				Marking details	Marks available					
					AO1	**AO2**	**AO3**	**Total**	**Maths**	**Prac**
	(c)	(i)		 (1) 1:1 molar ratio 0.150 × 159.8 = 23.97 g (1)		2		2		
		(ii)		potassium cyanide / sodium cyanide	1					1
		(iii)		dilute sulfuric acid / hydrochloric acid	1					1
				Question 13 total	**5**	**5**	**4**	**14**	**0**	**3**